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18 December 1990
SUPERSEDING
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15 February 1973

MILITARY SPECIFICATION

Milling Machines, Bed Type, Vertical, Double Housing, Numerically Controlled, Profiling, 5 to 25 Horsepower Per Spindle, Single and Multi-Spindle Machines.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers numerically controlled, bed type, double housing vertical profilers with horsepower range from 5 to 25, table widths from 30 to 120 inches, and table lengths up to 312 inches, single and multiple motor-driven spindles with equipment and accessories specified in section 3. The requirements set forth in this specification and the completed data (paragraph 6.2) establish the Government's minimum requirements for each procurement.

1.2 Classification. The milling machines shall be of the following types, classes, and sizes. The type, class and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Single Spindle Machine.
Type II - Multi-Spindle Machine - each spindle assembly constructed as an individual unit and driven independently.
Type III - Multi-Spindle Machine - two or more spindles mounted in a single housing and driven by one motor through a common spindle drive.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type I machines shall include Class 1 only. Type II and III machines shall include Classes 2 through 4.

Class 1 - One spindle machine
Class 2 - Two spindle machine
Class 3 - Three spindle machine
Class 4 - Four spindle machine

Classes 1 and 2 shall include Sizes 1 through 10. Classes 3 and 4 shall include Sizes 3 through 10 only.

Size 1 - 30 inches minimum table width
Size 2 - 36 inches minimum table width
Size 3 - 48 inches minimum table width
Size 4 - 60 inches minimum table width
Size 5 - 66 inches minimum table width
Size 6 - 72 inches minimum table width
Size 7 - 78 inches minimum table width
Size 8 - 84 inches minimum table width
Size 9 - 96 inches minimum table width
Size 10 - 120 inches minimum table width

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, ATTN: Mr. Stephen Lowell, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook - Tolerances and Measuring Methods for Unassembled Spur and Helical Gears (including Metric Requirements). (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. (DOD adopted)

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(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME B5.1M	- T-Slots, Their Bolts, Nuts, and Tongues, and Cutters. (DOD adopted)
ANSI B5.18	- Spindle Noses and Tool Shanks for Milling Machines. (DOD adopted)
ANSI B11.8	- Drilling, Milling, and Boring Machines, Safety Requirements for the Construction, Care and Use of. (DOD Adopted)
ANSI B93.114M	- Pneumatic Fluid Power Systems Standard for Industrial Machinery. (DOD adopted)
ISO 54	- Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.
ANSI/ISO 8860/1	- Dimensional, Physical, and Magnetic Characteristics.
ANSI/ISO 8860/2	- Track Format.
ANSI/ISO 9293	- Volume and File Structure of Flexible Disk Cartridges for Information Interchange.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227	- One Inch Perforated Tape. (DOD adopted)
RS-232	- Data Terminal Equipment and Data Communication Equipment, Interface Between, Employing Serial Binary Data Interchange. (DOD adopted)
RS-244	- Character Code for Numerical Machine Tool Control Perforated Tape. (DOD adopted)
RS-267	- Axis and Motion Nomenclature for Numerically Controlled Machines. (DOD adopted)
RS-274	- Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines. (DOD adopted)

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RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DOD adopted)

RS-422 - Electrical Characteristics of Balanced Voltage Digital Interface Circuits.

RS-449 - General Purpose 37-Position and 9-Position Interface for Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange.

RS-494 - 32 Bit Binary CL (BCL) Exchange Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006-1899.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes

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precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The milling machine shall be proven (not a prototype) and one of the manufacturer's new current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the mill that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the

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operation of the machine. All guards shall allow access to the guarded part(s).

3.2.6 Environmental protection. The machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. During the operation of the machine at its final location, it shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract regulating the emission of materials hazardous to the environment. All fluids and greases initially used charging the machine shall also comply with applicable EPA and OSHA restrictions and shall not be hazardous to the environment or to the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations as required herein shall be as specified by the procuring activity (see 6.2).

3.2.7 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.8 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. Replacement parts shall be domestically available.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent unintentional change of tightness. Those subject to removal or adjust shall not be swaged, peened, welded, staked, or otherwise, permanently deformed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any excess material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations. None of these operations shall be employed as a repair measure for defective parts.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2001-B88 and AGMA 2000-A88 for the inch-pound system of measurement or ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The milling machine shall consist of not less than a bed, a table, housings, bridge rail, spindle(s) saddle, spindle headstock, spindle assembly and a computer control system.

3.4.1 Bed. The bed shall be a casting of iron alloy or fabricated from steel. The bed shall contain the feed drive mechanism for the table and have the necessary access doors or covered openings to facilitate required adjustments and replacement of parts. The length of the bed and bed ways shall be such that the worktable is fully supported and does not overhang the bed in any position within its rated travel. Leveling screws and holddown bolts shall be provided to enable the machine to be properly installed and leveled.

3.4.2 Table. The table shall be a casting of iron, iron alloy or steel. The table shall be supported on anti-friction roller bearings on replaceable way strips fitted on the bedways with means to adjust the ways. The location of the table on the bed shall be such that the table will be fully supported throughout its range of travel. Unless otherwise specified (see 6.2), the table shall have not less than one T-slot conforming to ANSI B5.1 every five inches, parallel to the longitudinal travel of the table. The table shall

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have coolant drain channels for returning the coolant to the reservoir with provision to prevent chips from entering the reservoir. Means shall be provided to prevent overtravel of the table. The table shall be provided with power and rapid traverse for longitudinal movement.

3.4.3 Housings. Each housing shall be a casting of iron or iron alloy or a steel weldment. The housings shall adequately support the bridge rail and spindle assembly. The tops of the housings shall have provisions for mounting the bridge rail. The housings shall be provided with access doors as required for servicing the components located therein. The bottom of each housing shall be provided with holddown bolts as well as leveling adjustment screws located to assure and maintain level of machine under maximum load conditions.

3.4.4 Bridge rail. The bridge rail shall be designed and constructed for rigidity, stability and minimum sag or deflection. The bridge shall be mounted on the top of the vertical housings parallel to the table surface. The bridge shall have hardened and ground ways of large proportions to provide the spindle and saddle with maximum support.

3.4.5 Spindle(s) saddle. The saddle, for the purposes of this specification, is defined as the structural part of the machine that moves along the ways of the fixed bridge rail parallel to the table surface and provides ways for the spindle(s) headstock to travel in a plane perpendicular to the table surface. The saddle shall be constructed for strength and rigidity to assure accurate vibration free performance within the full range and capacity of the machine.

3.4.6 Spindle headstock. The headstock, carrying one or more spindles, shall be a vertical sliding structure mounted on the saddle ways, adequately proportioned to provide sufficient rigidity of the machine. The headstock shall be supported to the saddle by way strips or anti-friction roller bearings. Simple means of adjustment to insure correct alignment of the mating parts shall be furnished.

3.4.7 Spindle assembly. Each spindle shall be made of heat treated alloy steel. When multiple spindles are required, the spindles shall be identical and shall be mounted to the headstock in line parallel to the table working surface. Spindle bearings shall be designed and assembled to compensate for thermal expansion. Means for maintaining a controlled temperature in the headstock spindle shall be provided. The spindle shall have precision ground Number 50 American National Standard Taper and shall be in accordance with ANSI B5.18. Spindle(s) shall have forward and reverse directions of rotation, controllable by electro-manual controls and the numerical control unit. The spindle(s) shall have features to alert the operator and cause all movements to stop if any spindle becomes overloaded, stalled, or overheated. Braking devices shall be interlocked to prevent power rotation of spindle(s) until the brake device has been released. Means shall be provided for selection of spindle speeds by electro-manual controls and numerical control. When dual horsepower spindle motors are required (see 6.2), the lowest horsepower shall not be less than that stated in table I for size machine ordered.

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3.4.8 Feed drive. The X, Y, and Z axes of the machine shall be actuated through a closed loop, high gain electric or electro-hydraulic servo-mechanism driving recirculating anti-backlash ball screws and pre-loaded recirculating ball screw nuts or rack and pinion drives. Infinitely variable feed drives shall be provided for all axes, actuated and controlled by commands from the control tape and by manually inserting data.

3.4.9 Power rapid traverse. Power rapid traverse shall be provided for movement of X, Y, and Z slides. The rate of rapid traverse shall be independent of any feed adjustment.

3.4.10 Way wipers and covers. The X and Y axes ways shall be provided with wipers and covers to protect the way bearing surfaces from dust and other foreign matter. The Z ways shall be provided with wipers or covers.

3.4.11 Computer numerical control (CNC). The machine shall have a fully automatic, solid-state, soft-wired, integrated circuit type, computer numerical control. Control interfaces, (EIA RS-232, EIA RS-449 or both) shall be provided as specified (see 6.2). When required, interface circuitry shall conform to EIA RS-422 (see 6.2). The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, linear axes movement, spindle operation, and other part program directed functions for machines of the type specified herein. Control axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of three linear axes. Control features shall include an EIA RS-232 interface, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory, manually input or directly from the input media specified herein. The control shall initiate a halt or an error message should a fault condition occur in the control. All necessary executive program routines shall be furnished. A line voltage of $\pm$10 percent from normal shall not adversely affect the numerical control system function. The control shall function in ambient temperatures ranging from 30 to 120° Fahrenheit, and humidity ranging from 5 to 95 percent non-condensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. The control shall be inch/metric switchable with control resolution of 0.0001 in/0.001mm for each axis.

3.4.11.1 Data input.

3.4.11.1.1 Punched tape data input. When specified (see 6.2), the input media for the control shall be 1-inch, 8-channel punched tape. The data input format and codes shall comply with EIA Standards RS-227, RS-274, and RS-358. When specified (see 6.2), RS-244 shall be used in lieu of RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, feeding, and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2), the tape reader shall be supplied with standard feed and take-up spools 5-1/4

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inches in diameter. The tape reader shall be housed in a dust free area of the numerical control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244.
b. Even parity per EIA RS-358.
c. Misaligned sprocket holes.
d. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.11.1.2 Binary cutter location. When 32 Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machines is specified (see 6.2), the input media shall be a 3-1/2 inch hardshell diskette structured in accordance with ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. The input data format shall comply with EIA Standard RS-494. Manufacturers providing functions or features that cannot be addressed by command codes currently covered by EIA RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features or functions. The control shall be equipped with a 3-1/2 inch hardshell diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.11.1.3 Diskette input. When specified (see 6.2), a portable diskette reader/writer shall be provided with all the hardware and software necessary to interface between the control and the reader/writer. The reader/writer shall receive and store part program data from the control unit and be capable of loading stored programs to the memory of the control unit. Part program storage shall be on a 3-1/2 inch hardshell diskette. The diskette shall meet the requirements of ANSI/ISO 8860/1, ANSI/ISO 8860/2, and ANSI/ISO 9293. The communication between the reader/writer shall be by means of EIA RS-232 communication ports.

3.4.11.1.4 Manual data input. A means for manual data input (MDI) shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.11.2 Postprocessor. When specified (see 6.2), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted program preparation. The postprocessor shall be provided for converting the specified program language (see 6.2) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2).

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3.4.12 Standard control features.

3.4.12.1 Block delete. The block delete feature shall permit bypassing any blocks of information identified by a block delete code for use when a portion of a program is to be omitted.

3.4.12.2 Buffer storage. The MCU shall have buffer storage capable of storing one or more blocks of program data read either from tape, diskette, or memory. As blocks are transferred to active command, new blocks shall be read into the buffer.

3.4.12.3 Display. Unless otherwise specified (see 6.2), the display shall be a cathode ray tube (CRT). The CRT shall be capable of displaying machine system parameters and program codes from memory, buffer storage, and active data. The CRT shall be capable of displaying 256 characters or more.

3.4.12.4 Program data edit. Part program edit capability shall be provided as means for editing tape and MDI input to optimize part program data. It shall be possible to modify part program data by adding, deleting, correcting, or completely changing stored data and operating with the edited program.

3.4.12.5 Sequence number search. The operator shall be able to search through the part program data for a designated block number by initiating the search through the use of the MDI keyboard. The search may be conducted through the tape data or stored part program data depending on which is active. Upon reaching the desired sequence number, it shall be possible to continue the program.

3.4.12.6 Jog. The jog control shall permit axis slide movement in the plus or minus direction of travel at preset jog rates such as high, low, or selectable rates of jog.

3.4.12.7 Diagnostic capability. The control shall include a diagnostic system for monitoring the control during operation and for locating malfunctions when troubleshooting. It shall bring the system to a controlled stop if a malfunction occurs during operation. The system shall give fault data in understandable code or message form on the display unit which will identify a problem at board level in the control and identify program errors.

3.4.12.8 Feedrate override. Means shall be provided to override the programmed feedrate from 0 to not less than 120 percent.

3.4.12.9 Programmed and optional stops. The control shall recognize programmed stopping points in the machining cycle. All controlled motions shall be stopped when the code for a programmed stop is read. The optional stop code shall be ignored unless the control is set to recognize the command.

3.4.12.10 Feed hold. The feed hold function shall permit the operator to interrupt feed rate and stop all motion. While in the feed hold condition,

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the machine may be operated manually. When normal operation is resumed, motions will resume at their previous feed rate.

3.4.12.11 Zero reference. Means shall be provided to establish a zero point at any position throughout the range of departure of each controlled axis. An operator control shall be provided to return the machine slides to the initial zero position for synchronizing the machine with the control.

3.4.12.12 Incremental feed. Incremental feed circuitry shall provide means for manually adjusting each slide position. Movements in the plus and minus directions shall be selectable in increments of 0.0001, 0.001, 0.01, 0.1 and 1.0 inch on each axis. The incremental feed shall function in each mode of operation except the automatic mode.

3.4.12.13 Feedrate programming. Feedrate shall be programmed in either inches per minutes (IPM) or revolutions per minute (RPM). In the feed per revolution mode all axial feed movement shall be a function of the spindle rotation. Feed per minute shall be programmed directly in inches per minute.

3.4.12.14 Part program storage. Part program storage shall be either solid state, hard disk, or bubble type memory. The MCU shall be capable of storing in memory the equivalent of not less than 1000 feet of post processed part program tape.

3.4.12.15 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10% of programmed speed. The spindle speed override function shall control not less than the range or override between 60% and 100% of programmed speed. The spindle speed override control shall have a position that allows return of speed to that which it was programmed.

3.4.12.16 Tool length offset. The control shall have a tool length offset feature to permit adjustments to the programmed tool length by means of manual data input. The tool length offset shall have a range of not less than the following:

a. Inch system: $\pm$0.0001 inch to $\pm$1.0000 inch.
b. Metric system: $\pm$0.001mm to $\pm$25.400mm.

3.4.12.17 Cutter compensation. Cutter compensation shall be provided to compensate for cutter diameters differing from those utilized in programming. The cutter compensation shall have a range of not less than the following:

a. Inch system: $\pm$0.0001 inch to $\pm$1.0000 inch.
b. Metric system: $\pm$0.001mm to $\pm$25.400mm.

3.4.12.18 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

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3.4.12.19 Reversal error compensation. Reversal error compensation circuitry shall be provided for a selectable dimension correction to be added automatically to the command dimension to each linear axis reversal to compensate for mechanical lost motion.

3.4.12.20 Lead error compensation. Lead error compensation software shall be provided to permit compensating for the lead errors of each axis ball screw.

3.4.13 Optional control unit features. The control unit shall be equipped with the optional features specified in 3.4.13.1 through 3.4.13.5.

3.4.13.1 Decimal point programming. When specified (see 6.2), the control shall accept decimal point programming and shall recognize and interpret numbers in a part program that contains a decimal point.

3.4.13.2 Independent feedback. If an independent feedback measuring system is required to be furnished (see 6.2), the feedback measuring system shall function as an independent unit so that accuracy is not affected by other components. It shall be of a type in which there is no physical contact between stationary and moving elements of the measuring device.

3.4.13.3 Computer graphics. When specified (see 6.2), computer graphic capabilities shall be provided as required and as fully described by the procuring activity.

3.4.13.4 Conversational programming. When specified (see 6.2), conversational programming shall be provided as required and as fully described by the procuring activity.

3.4.13.5 Probing capabilities. When specified (see 6.2), the machine shall have probing capabilities as required, and as fully described by the procuring activity.

3.4.14 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area.

3.4.15 Pneumatic system. The pneumatic system shall conform to the requirements of ANSI B93.114.

3.4.16 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of

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clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate if the reservoir is not removable.

3.4.17 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 230/460-volts, 3-phase, 60-Hertz (Hz). Initial wiring of the milling machine shall be as specified (see 6.2).

3.4.17.1 Motors. All motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.18 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.19 Hour meter. When specified (see 6.2), the milling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the mill shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each mill shall meet the alignment tolerances specified in table II.

3.7 Performance. The machine shall be capable of satisfactory performance when tested in accordance with tests stated under paragraph 4.8.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

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3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified and fully described (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or machine.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE number or military or Federal specification number)
Type of filter

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3.12 Workmanship. The workmanship of the mill and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection

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shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each mill to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^0 to 104^0 Fahrenheit and 20 to 95 percent relative humidity.

4.8.2 Operational test. Each milling machine shall be operated under numerical control for a continuous period of 24 hours without load. The numerical control program shall cycle each axis through its entire range, addressing all speeds, feeds, direction of rotation, and standard control functions specified in this document. Proper operation of all machine functions, controls, motors, adjusting mechanisms, and accessories shall be verified during the test period. Should a malfunction or failure occur, the test shall be restarted at the beginning.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.8.4.1 through 4.8.4.5. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. The following tests shall be performed to test the functional accuracy, dynamic stiffness, and servo response of the machine and control under operating conditions. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.8.4.1 Metal removal and maximum horsepower test. The material for this test shall be SAE/AISI 4140 or 4340 annealed steel or equivalent, with a Brinell hardness of 300 to 350. Not less than 4 workpieces shall be used; thickness not less than 2.5 inches, with area not less than 144 square inches. The workpieces shall be placed on the worktable to cover a range requiring not less than 75 percent of the cross and longitudinal travel. The spindle speed and feed rates shall be such that the chip load is not less than 0.012 inch and the peripheral cutter speed shall not exceed 325 surface feet per minute

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(sfm). The depth of the cut shall be sufficient to load the spindle motor to its maximum rated horsepower (hp) of ±5 percent. The meter used in this test shall have an accuracy of not greater than ±2 percent. Except as limited herein, the cutter speed, feed rate, cutter type, and cutter diameter shall be selected by the manufacturer. The cutting action shall be smooth and even. The surface of the workpieces after the cut shall show no visual evidence of tool chatter, and the surface of the workpieces shall meet the following requirements:

Finish	150 roughness height rating (rhr)
Flatness	0.002 inch per foot, plus 0.0002 inch per foot accumulative

Metal removal shall be not less than the rated hp x 0.65 cubic inches per minute. With multi-spindle machines, the center spindle only shall be used for the test.

4.8.4.2 Overshoot and undercut test. The overshoot and undercut test shall be performed in accordance with paragraph 4.3.3.4.1 of NAS 979, sheet 19, copyrighted 1969, to test the dynamic servo response of the machine and control. The tolerance for the completed test piece shall not exceed the following:

Overshoot and undercut in X and Y axes; 0.0015 inch for Test A

Overshoot and undercut in X and Y axes; 0.002 inch for Test B

Overshoot in Z axis; 0.0015 inch for Test A

Overshoot in Z axis; 0.002 inch for Test B

4.8.4.3 Composite cutting test. The composite cutting test shall be performed in accordance with paragraph 4.3.3.5.1 of NAS 979, sheets 23 and 24, copyrighted 1969, to test the profiling and contouring capabilities of the machine and control unit. A feed rate of 25ipm and a spindle speed of approximately 1600 revolutions per minute (rpm) shall be used for this test. The tolerances for the completed test piece shall be as follows:

Squareness	0.002 inch Total Indicator Reading (TIR)
Parallelism	0.002 inch TIR
Angularity	0.00025 inch TIR (5° ramp)
Circle	±0.00025 inch on diameter
Flatness	0.002 inch TIR
Finish	32 rhr, all surfaces

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4.8.4.4 Positioning accuracy and repeatability. The positioning accuracy and repeatability of the machine slides shall be evaluated in accordance with the National Machine Tool Builders' Associations Definition and Evaluation of Accuracy and Repeatability for Numerically Controlled Machine Tools, Section 1 - Linear Definition. At least 10 random target points, spaced over the full travel of each axis shall be selected and approached from both directions.

4.8.4.5 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The milling machines covered by this specification are to be used in the shops and depots of activities in the Department of Defense.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound or metric) (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

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g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility requirements (see 3.3.8).

j. T-slots, if different (see 3.4.2).

k. Dual horsepower spindle motors, if required (see 3.4.7).

l. Specify interface configuration (EIA RS-232, EIA RS-449 or both) (see 3.4.11).

m. Specify EIA RS-422 interface circuitry, if required (see 3.4.11).

n. Specify punched tape input, if required (see 3.4.11.1.1).

o. Specify EIA Standard RS-244 in lieu of EIA RS-358, if required (see 3.4.11.1.1).

p. Specify take up reels, if different (see 3.4.11.1.1).

q. Specify BCL, if required (see 3.4.11.1.2).

r. Specify diskette input, if required (see 3.4.11.1.3).

s. Postprocessor, if required (see 3.4.11.2).

t. Specify program language, if postprocessor is required, (see 3.4.11.2).

u. Specify computer type, if postprocessor is required, (see 3.4.11.2).

v. Specify display, if different (see 3.4.12.3).

w. Decimal point programming, if required (see 3.4.13.1).

x. Independent feedback, if required (see 3.4.13.2).

y. Computer graphics, if required (see 3.4.13.3).

z. Conversational programming, if required (see 3.4 13.4).

aa. Probing capabilities, if required (see 3.4.13.5).

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bb. Electrical requirements, if different (see 3.4.17).

cc. Specify initial voltage required (see 3.4.17).

dd. Motors, if different (see 3.4.17.1).

ee. Worklight, if required (see 3.4.18).

ff. Hour meter, if required (see 3.4.19).

gg. Size and capacity, if different (see 3.5).

hh. Specify physical size limitation, if required (see 3.5.1).

ii. Additional performance features, if required (see 3.7.1).

jj. Specify and fully describe optional and accessory equipment, if required (see 3.9).

kk. Specify applicable repair parts, if required (see 3.10).

ll. Lubrication plate, if required (see 3.11.1).

mm. Information to be provided on lubrication plate, if different (see 3.11.1).

nn. Nameplate, if different (see 3.11.2).

oo. Hydraulic plate, if required (see 3.11.3).

pp. Hydraulic plate informational, if different (see 3.11.3).

qq. Inspection conditions, if different (see 4.3).

rr. First article inspection, if different (see 4.4).

ss. Quality conformance inspection, if different (see 4.5).

tt. Acceptance test, if different (see 4.6).

uu. Additional performance tests, if required (see 4.8.4.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be

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prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
6.3.1	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language.
6.3.1	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, and machine alignment and accuracy test results) ordered shall be written in the English language. See paragraph 6.3 for additional information.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268.

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6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

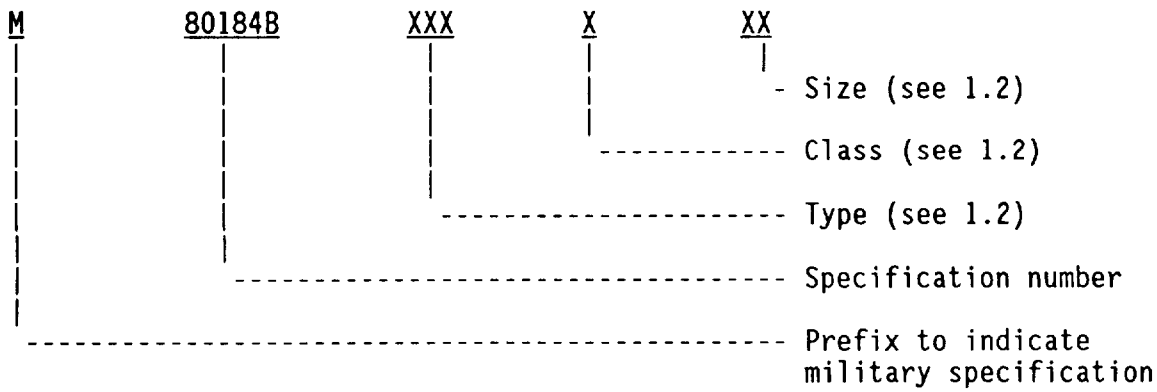
6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. All training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. All warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Assembly
Bridge rail
Housings
Saddle

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TABLE I. Sizes and Capacities.

CHARACTERISTIC	Size 1	Size 2	Size 3	Size 4	Size 5
Table size, working surface					
Width, inches	30	36	48	60	66
Length, inches	36	48	48	72	96
Optional lengths, inches	48, 60	60, 72	72,96, 120	96,120, 144	120,144
Feed range, inches					
Longitudinal (X)	NOT LESS THAN TABLE LENGTH				
Cross (Y) single spindle	AT LEAST 4 INCHES MORE THAN TABLE WIDTH				
Cross (Y) multi-spindle	NOT LESS THAN DISTANCE BETWEEN SPINDLES				
Vertical (Z)	12	14	14	14	18
Quill movement or depth adjustment	1	1	1	1	1
Spindle nose to table, inches	18	18	18	18	18
Feed rates, number of	INFINITE				
Feed rate, range, IPM,					
Longitudinal (X)	.5 TO 100	.5 TO 100	0 TO 150	0 TO 150	0 TO 150
Cross (Y)	.5 TO 100	.5 TO 100	0 TO 150	0 TO 150	0 TO 150
Vertical (Z)	.5 TO 100	.5 TO 100	0 TO 150	0 TO 150	0 TO 150
T-slots size	13/16	13/16	13/16	13/16	13/16
T-slots center to center distance	5	5	5	5	5
Number of spindles	1	1	1	1	1
Optional number of spindles	2	2	2 OR 3	2, 3 OR 4	2, 3 OR 4
Distance between spindles					
2 spindles	12	15	20	30	33
3 spindles	-	-	15	18	20
4 spindles	-	-	-	15	16
Spindle motor, horsepower per spindle	5	5	5	5	5
Optional horsepower per spindle	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25
Spindle speeds, number of	INFINITE				
Optional number of speeds	24	24	24	24	24
Spindle speed range,RPM	36-3600	36-3600	36-3600	36-3600	36-3600
Optional range	24-2400 or 18-1800	24-2400 or 18-1800	24-2400 or 18-1800	24-2400 or 18-1800	24-2400 or 18-1800
Spindle horsepower per RPM	1/4	1/4	1/4	1/4	1/4
Optional HP/RPM	1/10	1/10	1/10	1/10	1/10
Distance between housings, inches	42	48	60	72	78
Power rapid traverse, IPM					
Longitudinal (X)	100	100	150	150	150
Cross (Y)	100	100	150	150	150
Vertical (Z)	100	100	150	150	150

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TABLE I. Sizes and Capacities - continued.

CHARACTERISTIC	Size 6	Size 7	Size 8	Size 9	Size 10
Table size, working surface					
Width, inches	72	78	84	96	120
Length, inches	96	96	96	120	144
Optional lengths, inches	120, 144	120, 144	144, 168, 192, 214	144, 168, 192, 214	192, 214, 240, 312
Feed range, inches					
Longitudinal (X)	NOT LESS THAN TABLE LENGTH				
Cross (Y) single spindle	AT LEAST 4 INCHES MORE THAN TABLE WIDTH				
Cross (Y) multi-spindle	NOT LESS THAN DISTANCE BETWEEN SPINDLES				
Vertical (Z)	18	18	18	18	18
Quill movement or depth adjustment	1	4	4	4	4
Spindle nose to table, inches	18	18	18	18	18
Feed rates, number of	INFINITE				
Feed rate, range, IPM,					
Longitudinal (X)	0 TO 150	0 TO 150	0 TO 150	0 TO 150	0 TO 150
Cross (Y)	0 TO 150	0 TO 150	0 TO 150	0 TO 150	0 TO 150
Vertical (Z)	0 TO 150	0 TO 150	0 TO 150	0 TO 150	0 TO 150
T-slots size	13/16	13/16	13/16	13/16	13/16
T-slots center to center distance	5	5	5	5	5
Number of spindles	1	1	1	1	1
Optional number of spindles	2, 3 or 4	2, 3 or 4	2, 3 or 4	2, 3 of 4	2, 3 or 4
Distance between spindles					
2 spindles	36	38	42	48	60
3 spindles	22	24	30	30	30
4 spindles	18	19	21	24	30
Spindle motor, horsepower per spindle	5	5	5	5	5
Optional horsepower per spindle	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25	7 1/2, 10 15, 20 or 25
Spindle speeds, number of	INFINITE				
Optional number of speeds	24	24	24	24	24
Spindle speed range	36-3600	36-3600	36-3600	36-3600	36-3600
Optional range	24-2400 or 18-1800	24-2400 or 18-1800	24-2400 or 18-1800	24-2400 or 18-1800	24-2400 or 18-1800
Spindle horsepower per RPM	1/4	1/4	1/4	1/4	1/4
Optional HP/RPM	1/10	1/10	1/10	1/10	1/10
Distance between housings, inches	84	90	96	108	132
Power rapid traverse, IPM					
Longitudinal (X)	150	150	150	150	150
Cross (Y)	150	150	150	150	150
Vertical (Z)	150	150	150	150	150

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TABLE II.

TEST	TOLERANCES
Spindle(s) periphery runout	0.0004 inch TIR
Spindle(s) face runout	0.0003 inch TIR
Spindle(s) internal taper runout	
1 inch from spindle(s) nose	0.0004 inch
12 inches from spindle(s) nose	0.001 inch
Longitudinal rise and fall of table	0.0005 inch in any 12-inch span Total not to exceed 0.0005 ±0.00015 per 12 inches of travel in excess of 12 inches
Crossrail ways parallel with table surface	0.0005 inch in any 12-inch span Total not to exceed 0.0005 ±0.00015 per 12 inches of travel in excess of 12 inches
Spindle(s) square with table (center of rotation)	0.001 per foot
Spindle(s) travel square with table	0.0005 inch in any 12-inch span Total not to exceed 0.0005 ±0.00015 per 12 inches of travel in excess of 12 inches
T-slots parallel with table travel	0.0005 inch in any 12-inch span Total not to exceed 0.0005 ±0.00015 per 12 inches of travel in excess of 12 inches
Spindle centerlines with respect to each other	±0.002 inch

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR
Navy - YD
Air Force - 84

User activities:
Army - AL
Navy - AS, OS, MC

Preparing activity:
DLA - IP

Project (3417-0189)

*U.S. GOVERNMENT PRINTING OFFICE: 1991--504-034/50087

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-80184B	21 Dec 90

3. DOCUMENT TITLE
Milling Machines, Bed Type, Vertical, Double Housing, Numerically Controlled, Profiling

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM	901-775-4791	683-4791
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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